

Appendix 3

Vulnerability coding scheme

Vulnerability	Definition	Examples
Authentication attack	This code is used for general descriptions of attacks that either break authentication and/or exploit authentication features of a system to break other security requirements.	"We found that the authentication mechanism could be exploited to break privacy."
Ballot reordering attack	This code is used for an adversary to manipulate the voting sequence to ensure a discarded or overwritten vote is counted instead of the voter's final, intended choice. This exploit targets systems that allow re-voting, where the rule is typically that only the last ballot cast is valid. (Baloglu et al., 2021 "PROVABLY IMPROVING...", p. 7	"Systems that allow re-voting are vulnerable to ballot reordering attacks."
Network attack	This code is used for exploiting unauthorized access, stealing data, or damage systems. The attack could originate from external or internal actors (Aljabri et al., 2021, p. 4).	"The system was found to be vulnerable to network attacks from both internal and external actors."
Hash-based attack	This code is used for attacks on hashes. A hash-based attack occurs when an adversary exploits a hash function that lacks collision resistance. [66, p. 168]	"Schemes that lack collision resistance are susceptible to hash-based attacks."
Quantum attacks	This code is used for all mention of attacks enabled by quantum computers, including e.g. brute	"... using a quantum computer even these cryptographic schemes could be broken via brute-forcing."

	forcing attacks specifically possible given quantum technology.	
Brute-forcing attack	This code is used for an adversary gaining physical access to a voter's smart card and systematically attempting to guess the PIN to cast a fraudulent vote. Because PIN codes must be short enough for humans to remember, the search space is limited, making it possible for an attacker to automate the process and eventually find the correct combination. (Estaji et al., 2020, p. 54)	"The short PIN requirement makes the system vulnerable to brute-forcing attacks."
Coercion attack	This code is used for general mentions of coercion. The code also includes some named attacks that were mentioned infrequently only once where the main aim of the attack is to enable coercion. Examples of attacks that fall under the category of coercion attack: vote buying [24], forced-abstention attacks, randomization attacks, and simulation attacks (Heinl et al., 2022, p.9).	"We found that the system was vulnerable to coercion attacks, including vote buying." "These types of schemes are still vulnerable to forced-abstention attacks." "... but this property does not defend against randomization or simulation attacks. To protect the system against these types of attacks..."
Ballot stuffing attack	This code is used for an adversary to cast a false vote using a valid voter identity. This attack directly targets the property of Result Integrity. (Baloglu et al., 2021 "PROVABLY IMPROVING...", p. 6)	"This means the system is vulnerable to ballot stuffing..."

Ballot replay attack	This code allows an attacker to capture an encrypted vote or a login token, and then send it again to the recipient to perform an unauthorized action (Babenko & Pisarev, 2021, p. 2)	"Systems that do not invalidate previously cast votes are susceptible to ballot replay attacks."
Poisoning attack	This code enables an attack where a malicious voter submits a fake or corrupted ballot that is mathematically flawed. This type of attack doesn't try to change votes but instead breaks the system's ability to count any votes (Chang-Fong & Essex, 2016, p. 4)	"An unsatisfied voter could send in a corrupted ballot which could make decryption of the results impossible." "... this is vulnerable to attacks aiming at invalidating the vote using malicious ballots."
Denial-of-service attack	This code is used where an attacker floods a voting or registration server with fake traffic. This malicious overload causes the server to slow down or crash. (Toapanta et al., 2020, p. 114)	"DoS attacks are still a problem for most blockchain systems in the literature, threatening availability." "One attack which can disrupt the election is a Denial of Service attack"
Single point of failure	This code is used for attacks that rely on a design flaw where there is a trusted component in the system which trivializes an attack on the system. For example, Estaji et al., 2020, describes a protocol that utilizes smart cards for voting which has the security assumption that the smart card is trusted and not modified to be malicious (p. 57).	"The reliance on a trusted smart card introduces a single point of failure in the system." "To avoid a single point of failure the system utilizes..." "This presents a single point of failure where the adversary only needs to guess the pin, which is easy to do using bruteforcing"

Receipt attack	This code describes attacks where coerced voters are forced to vote in a way where the coercer can verify that they followed their instruction. This includes the italian attack, where the voter is instructed to vote with a pattern (e.g. some kind of personal identifier) that makes whether they followed the coercers instructions verifiable [13]. We also include some cases of ballot copying attacks [24], since the coercer can verify whether the voter has followed their instructions.	"This scheme is vulnerable to the Italian attack." "This is vulnerable to ballot copying..." "This scheme is vulnerable to receipt attacks"
Defaming attack	This code is used for attacks where the adversary pretends that a system error has occurred by e.g. fabricating a vote receipt and then claiming it was cast but was omitted from the Web Bulleting Board (Culnane et al., 2015, p. 8).	"By fabricating a vote receipt, an adversary could perform a defaming attack against the election authority."
Kleptographic attack	This code is used to describe kleptographic attacks, where a backdoor is inserted into e.g. a cryptographic algorithm that allows an adversary to steal confidential information (Sajjad et al., 2020).	"Schemes that do not protect against backdoors are susceptible to kleptographic attacks."
Sybil attack	This code is used for attacks where the voter attempts to gain access to multiple identities to modify the election (Bartolucci et al, 2018, p. 31). Examples include making multiple pseudonyms (i.e. public	"The blockchain system is vulnerable to a Sybil attack, as you can always generate multiple public keys to sign up with"

	keys) on the blockchain to vote (Ibid.).	
Malware	This code is used for attacks that use malware to infect a voter server. The attacker can then compromise the server and system and damage vital information.	"We found that the voting server was vulnerable to malware, compromising stored ballots."
Social engineering	This code is used for a variety of attacks with the common factor that they aim at exploiting human factors in the system (Hatfield, 2018). This code includes phishing (Khonji, 2013), impersonation of the voter and/or election authority, and some types of spoofing (Hatfield, 2018, p. 107).	"Fraud polling stations is a problem we have seen in..." "Voters could be exposed to phishing were the adversary impersonates an election authority..."
Injection attack	This code is used for vulnerabilities like integer overflows, SQL Injections which manipulate database queries to corrupt data, and XSS which inserts malicious scripts to gain authorized access or modify database content.	"We found that the system was vulnerable to SQL injection, allowing an attacker to manipulate the voting database." "In this study we show that the system is vulnerable to a Cross-Site-Scripting (XSS) attack..." "In our literature review we found the following vulnerabilities in e-voting systems: ballot copying attack, social engineering, inter overflow, man-in-the-midde attacks..."

Insider attack	This code is used for exploits that are characterized by being used by an insider in the system. An insider could be a corrupt election authority or a secret share holder in a system using, for example, Shamir's Secret Sharing [66, p.541].	"The system is susceptible to a last voter's share attack." "A corrupt election authority could potentially publish the results partway."
Tamper attack • tamper attack (other) • tamper attack (ballot)	This code is used for general mentions of tampering as well as cases where the ballot is being tampered with. If it is not clear what the object that is being tampered with is then code "tamper attack (other)". If tampering is specifically directed at modifying the ballot, then we use the subcode "tamper attack (ballot)". If hardware is tampered with, use the "hardware exploitation" code instead.	"... allowing us to tamper with the ballots" - code "tamper attack (ballot)" since it specifically mentions the ballot is tampered with. "The system was found to be vulnerable to tampering..."
Ballot design manipulation	This code is used for attacks where the attacker can change the ballot options, e.g. the candidates or the voter's options in terms of what they can select to manipulate the outcome (Specter and Halderman, 2021, p.8-10).	"This scheme is vulnerable to ballot design manipulation, allowing an attacker to modify the list of candidates."
Ballot misdirection attack	This code is used for attacks where the return instructions of the ballot are modified, causing the voted ballots to be sent to the wrong sorting location or delayed until	"We found that we could change the return instructions of the ballot, which could be exploited to delay the vote, potentially until after the election."

	after the vote (Specter and Halderman, 2021, p.8).	
Forgery attack	This code is used for mentions of forgery as an attack on a system, this includes credential forgery and vote forgery. When a more specific subcode better explains the exploit used such as “ballot replay attack”, then we code that instead.	"The scheme was found to be vulnerable to forgery attacks, allowing an adversary to forge voter credentials."
Hardware exploitation	This code is used for all mentions of exploiting hardware for elections. This includes mentions of tampering where the target is hardware, or an exploit that specifically targets a physical device.	"We found that we could tamper with the screen contents of the DRE." "It was possible to hack into the DRE and upload malware".
Clash attack	This code is used for all mentions of clash attacks, attacks where a voting machine attempts to provide different voters the same voting receipt, allowing election authorities to replace the identical receipts with new ballots affecting the election (Kusters et al., 2012).	"This scheme is vulnerable to clash attacks, as the voting machine can issue identical receipts to different voters."
Man-in-the-middle attack	This code is used for mentions of man-in-the-middle attacks, attacks where an adversary intercepts communications between two parties with the aim of stealing	"In our literature review we found the following vulnerabilities in e-voting systems: ballot copying attack, social engineering, inter overflow, man-in-the-middle attacks..."

	confidential information (Regenscheid, n.d.)	
Timing attack	This code is used for mentions of timing attacks, attacks that aim to gain confidential information about e.g. a decryption algorithm by timing how long it takes to perform certain steps (Crosby et al., 2009). This includes direct mentions of timing attacks as well as described attacks that utilize the premise of timing algorithms to gain information used for an exploit.	"This decryption scheme is susceptible to timing attacks, potentially leaking information about private keys."
Zero-day attack	This code is used for mentions of zero-day attacks, attacks that are unknown to the developers (Ahmad et al., 2023). We only coded "zero-day attack" if the paper used the literal term.	"One of the identified problems in the literature was zero-day attacks..."